

**COLD LAKE, AB, Canada**

WMO: 711200

Lat: **54.420N**Lon: **110.280W**Elev: **541**StdP: **94.99**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -33.0      | -30.0 | -37.9                         | 0.1 | -32.9 | -35.0 | 0.1 | -29.9 | 11.0                  | -2.7 | 9.3 | -5.6 | 1.4                   | 230  | 0.660 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 28.1            | 18.1 | 26.4 | 17.1 | 24.8 | 16.3 | 19.3                | 26.2 | 18.2 | 24.5 | 17.2 | 23.0 | 3.9                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.7                            | 12.7 | 22.3 | 15.7 | 11.9 | 21.2 | 14.9 | 11.3 | 20.0 | 57.4          | 26.3 | 53.9 | 24.5 | 50.6 | 23.1 | 23.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.3           | 8.1  | 7.1 | -36.7                      | 31.4 | 3.3                | 1.9 | -39.1                                              | 32.8 | -41.1      | 34.0 | -42.9      | 35.0 | -45.4      | 36.5 |    |    |
| (5)               |      |     | -36.8                      | 21.3 | 3.3                | 1.0 | -39.2                                              | 22.0 | -41.1      | 22.6 | -43.0      | 23.1 | -45.4      | 23.9 |    |    |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.4   | -14.6 | -12.1 | -5.3  | 3.5   | 10.5  | 15.1  | 17.8  | 16.3  | 11.1  | 3.9   | -5.7  | -13.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 13.00 | 8.92  | 7.73  | 7.81  | 5.54  | 4.64  | 3.35  | 2.89  | 3.34  | 4.37  | 4.73  | 6.46  | 8.00  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 3532  | 764   | 619   | 473   | 202   | 50    | 3     | 0     | 1     | 38    | 195   | 472   | 714   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5884  | 1022  | 852   | 732   | 444   | 244   | 106   | 44    | 80    | 218   | 448   | 722   | 972   | (9)  |
| (10) |                                                                                  | CDD10.0     | 746   | 0     | 0     | 0     | 8     | 66    | 156   | 242   | 196   | 72    | 6     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 57    | 0     | 0     | 0     | 0     | 2     | 9     | 28    | 17    | 2     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 693   | 0     | 0     | 0     | 2     | 55    | 112   | 282   | 198   | 43    | 1     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 127   | 0     | 0     | 0     | 0     | 11    | 19    | 54    | 36    | 7     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.1   | 2.7   | 2.8   | 3.3   | 3.7   | 3.6   | 3.2   | 3.1   | 2.9   | 3.2   | 3.3   | 3.0   | 2.7   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 464   | 21    | 16    | 19    | 23    | 42    | 72    | 88    | 74    | 45    | 17    | 22    | 23    | (15) |
| (16) |                                                                                  | PrecMax     | 652   | 42    | 40    | 54    | 63    | 101   | 182   | 186   | 171   | 101   | 51    | 60    | 62    | (16) |
| (17) |                                                                                  | PrecMin     | 330   | 9     | 2     | 1     | 2     | 4     | 8     | 40    | 8     | 11    | 4     | 2     | 4     | (17) |
| (18) |                                                                                  | PrecStd     | 89    | 8     | 9     | 12    | 13    | 26    | 45    | 36    | 37    | 27    | 12    | 12    | 12    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 5.4   | 6.2   | 11.7  | 22.5  | 28.0  | 28.9  | 30.5  | 30.1  | 27.6  | 20.7  | 9.7   | 5.2   | (19) |
| (20) |                                                                                  |             | MCWB  | 2.3   | 2.2   | 5.3   | 11.2  | 14.9  | 17.4  | 20.1  | 18.9  | 16.5  | 12.3  | 4.8   | 2.2   | (20) |
| (21) |                                                                                  | 2%          | DB    | 2.8   | 3.1   | 8.8   | 18.5  | 24.6  | 26.2  | 28.2  | 27.3  | 24.1  | 16.3  | 6.5   | 2.2   | (21) |
| (22) |                                                                                  |             | MCWB  | 0.7   | 0.3   | 3.8   | 8.9   | 13.6  | 16.1  | 18.9  | 18.1  | 15.3  | 9.5   | 3.1   | 0.1   | (22) |
| (23) |                                                                                  | 5%          | DB    | 0.5   | 0.7   | 6.6   | 15.5  | 22.0  | 24.3  | 26.4  | 25.5  | 21.7  | 13.6  | 4.3   | 0.1   | (23) |
| (24) |                                                                                  |             | MCWB  | -1.3  | -1.3  | 2.4   | 7.3   | 12.0  | 15.2  | 18.0  | 17.2  | 14.0  | 7.9   | 1.9   | -1.5  | (24) |
| (25) |                                                                                  | 10%         | DB    | -2.2  | -1.2  | 4.6   | 12.8  | 19.7  | 22.3  | 24.7  | 23.7  | 19.1  | 11.3  | 2.3   | -2.1  | (25) |
| (26) |                                                                                  |             | MCWB  | -3.5  | -2.8  | 1.2   | 6.1   | 10.8  | 14.1  | 17.0  | 16.4  | 12.7  | 6.7   | 0.2   | -3.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.7   | 2.6   | 6.0   | 11.7  | 15.9  | 18.9  | 21.1  | 20.2  | 17.5  | 12.8  | 5.5   | 2.5   | (27) |
| (28) |                                                                                  |             | MCDWB | 5.0   | 5.6   | 11.1  | 20.9  | 24.8  | 26.0  | 28.5  | 27.8  | 25.3  | 19.8  | 8.6   | 4.8   | (28) |
| (29) |                                                                                  | 2%          | WB    | 0.8   | 0.4   | 4.0   | 9.4   | 14.3  | 17.4  | 19.9  | 18.8  | 15.9  | 10.2  | 3.6   | 0.2   | (29) |
| (30) |                                                                                  |             | MCDWB | 2.8   | 2.6   | 8.3   | 17.7  | 22.8  | 23.8  | 26.9  | 25.8  | 23.0  | 15.0  | 6.1   | 1.9   | (30) |
| (31) |                                                                                  | 5%          | WB    | -1.3  | -1.1  | 2.7   | 7.7   | 12.9  | 16.2  | 18.8  | 17.9  | 14.4  | 8.4   | 2.0   | -1.5  | (31) |
| (32) |                                                                                  |             | MCDWB | 0.3   | 0.7   | 6.2   | 14.8  | 20.3  | 22.0  | 25.0  | 24.1  | 20.7  | 12.9  | 4.1   | 0.0   | (32) |
| (33) |                                                                                  | 10%         | WB    | -3.5  | -2.8  | 1.5   | 6.3   | 11.6  | 15.2  | 17.7  | 16.9  | 13.1  | 6.8   | 0.4   | -3.4  | (33) |
| (34) |                                                                                  |             | MCDWB | -2.2  | -1.2  | 4.3   | 12.2  | 18.2  | 20.5  | 23.2  | 22.5  | 18.1  | 10.7  | 2.2   | -2.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.7   | 10.2  | 10.3  | 10.8  | 12.0  | 11.0  | 11.2  | 11.1  | 9.0   | 7.2   | 7.9   | (35)  |      |
| (36) |                                                                                  |             | MCDBR | 10.6  | 12.3  | 11.8  | 15.6  | 16.0  | 14.1  | 13.7  | 14.5  | 15.8  | 13.8  | 9.5   | 9.8   | (36) |
| (37) |                                                                                  |             | MCWBR | 8.9   | 10.0  | 7.8   | 8.7   | 7.7   | 6.3   | 6.4   | 6.9   | 8.4   | 8.4   | 6.9   | 8.4   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 10.4  | 11.8  | 11.4  | 14.6  | 14.2  | 11.8  | 12.4  | 12.8  | 14.7  | 13.0  | 9.2   | 9.6   | (38) |
| (39) |                                                                                  |             | MCWBR | 8.9   | 9.8   | 7.8   | 8.4   | 7.2   | 6.0   | 6.1   | 6.4   | 8.0   | 8.2   | 7.0   | 8.3   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.246 | 0.275 | 0.279 | 0.328 | 0.346 | 0.364 | 0.363 | 0.367 | 0.312 | 0.278 | 0.248 | 0.218 | (40)  |      |
| (41) |                                                                                  | taud        | 2.305 | 2.259 | 2.393 | 2.397 | 2.397 | 2.356 | 2.409 | 2.355 | 2.516 | 2.582 | 2.459 | 2.340 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 732   | 820   | 912   | 902   | 900   | 883   | 877   | 849   | 866   | 832   | 742   | 707   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 54    | 81    | 89    | 103   | 109   | 115   | 107   | 107   | 80    | 59    | 46    | 41    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.86  | 1.90  | 3.43  | 4.66  | 5.67  | 5.78  | 5.88  | 4.80  | 3.36  | 1.88  | 0.86  | 0.60  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.09  | 0.15  | 0.24  | 0.38  | 0.48  | 0.41  | 0.33  | 0.38  | 0.36  | 0.31  | 0.11  | 0.06  | (45)  |      |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (36 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page